Lecture 11: Cognitive Architectures

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- **Today's Topics**

1- Cognitive architectures

- **Session Learning Outcomes**

By the end of today's lecture you will be able to:

1- Describe cognitive architectures and the different categories of architectures.

What are the characteristics of a cognitive agent?

The chief characteristic of a cognitive agent is the ability to act effectively in a world that is uncertain, under-specified, dynamic, possibly cooperating with other cognitive agents

To achieve goals adaptively and robustly in these circumstances requires a complex system that can

- Construct models of the way the world works,
- Use them to guide actions prospectively, and
- Update them dynamically as the system continually learns through its interactions

A cognitive architecture is the way we specify what is required to achieve this.

What is a cognitive architecture?

A cognitive architecture is a software framework that integrates all the elements required for a system to exhibit the characteristic attributes of a cognitive agent

The design of a cognitive architecture requires the specification of the formalisms for all the processes and knowledge representations used by that framework

How does a cognitive architecture work?

A cognitive architecture integrates the core cognitive abilities so that these abilities can be dynamically coordinated

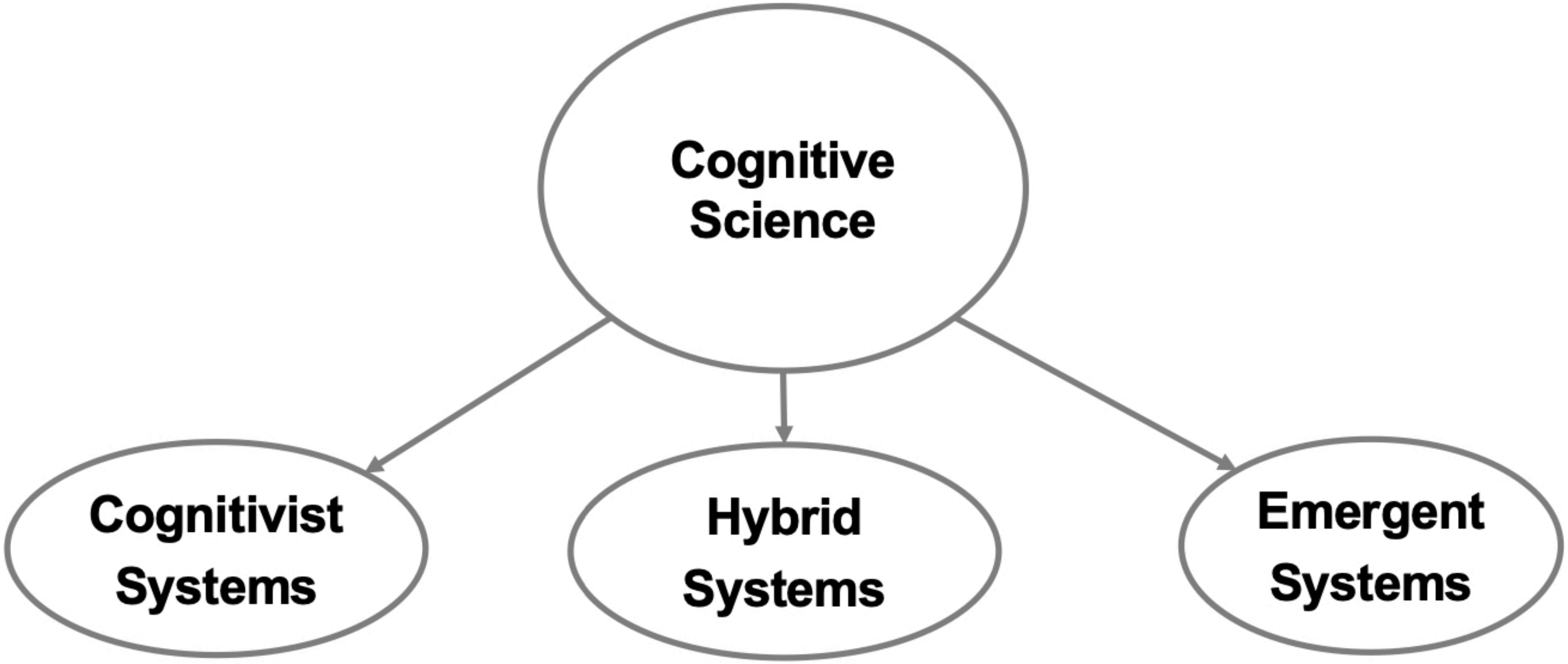
Perception
Attention
Action selection
Memory
Learning
Reasoning
Meta-reasoning
Prospection

Allowing the agent to exhibit flexible context-sensitive behaviour, prospectively selecting and controlling the actions that are required to achieve given goals

A cognitive architecture should also be able to develop autonomously so that its performance improves over time with experience

- **Prospection: Simulating future scenarios and predicting potential outcomes**

Cognitive Architectures



There are three paradigms of cognitive science

Cognitivist Cognitive Architecture

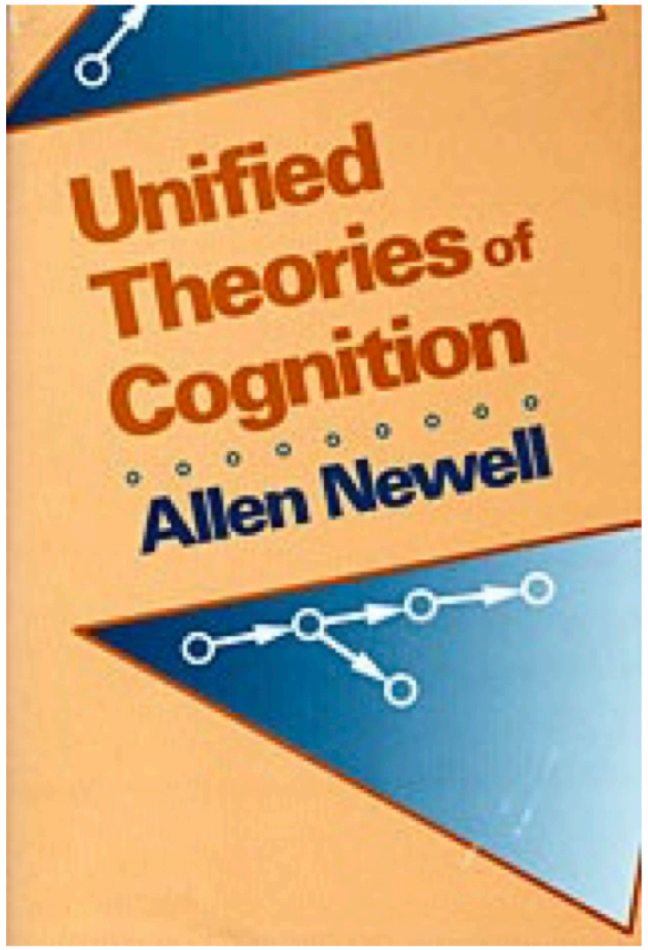
Attempts to create Unified Theories of Cognition (UTC)

UTCs cover a broad range of cognitive issues

- Attention
- Memory
- Problem solving
- Decision making
- Learning

from several aspects

- Psychology
- Neuroscience
- Computer Science



https://www.hup.harvard.edu/catalog.php?isbn=9780674921016

Cognitivist Cognitive Architecture

An encapsulation of a scientific hypothesis about those aspects of human cognition that are

- relatively constant over time and
- relatively independent of task

(Ritter and Young 2001)

Cognitivist Cognitive Architecture

- Generic computational model:
  - Not domain-specific
  - Not task-specific
- Knowledge provides the required specificity:

Cognitive Architecture + Knowledge = Cognitive Model

- Lehman et al. (1998) put it slightly differently:

BEHAVIOR = ARCHITECTURE x CONTENT

Content = the specific information that an agent is processing at any given moment

The effect of the architecture on behavior is not simply additive to the effect of the content

Cognitivist Cognitive Architecture

Knowledge is typically:

- Determined by the designer (explicitly or implicitly)
- Adapted and augmented by machine learning techniques

Cognitivist Cognitive Architecture

Overall structure and organization of a cognitive system

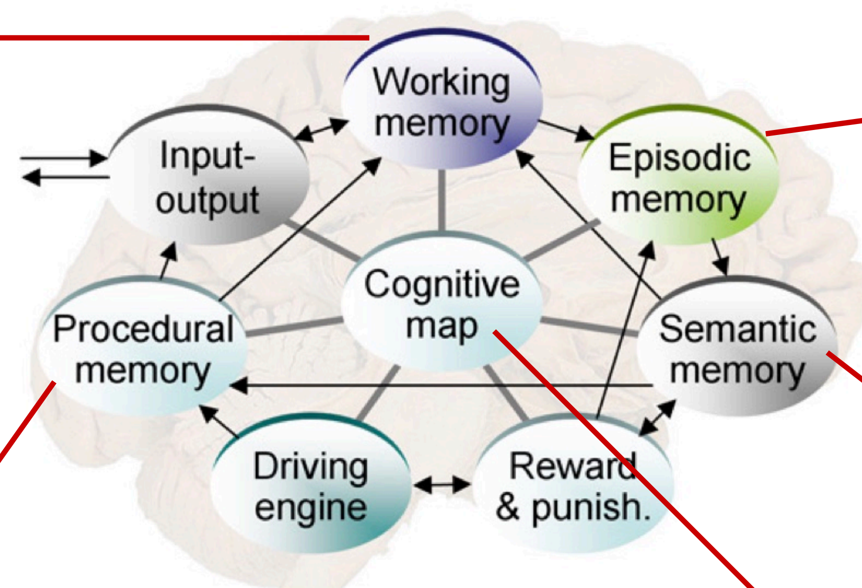
- Essential modules
- Essential relations between these modules
- Essential algorithmic and representational details in each module

[Sun 2007]

Short-term memory used in tasks like decision-making

Commitment to formalisms for representation and processes

(Langley 2005, Langley 2006, Langley et al. 2009)



GMU BICA Architecture (Samsonovich 2010)

Long-term memory encodes, stores, and retrieves specific episodes of the agent's past experiences

Knowledge, facts, and concepts

Sensory information

It models a variety of cognitive functions; e.g., spatial reasoning, planning, and navigation

Emergent Cognitive Architecture

Emergent approaches focus on development

- From a primitive state
- To fully cognitive state, over the system's lifetime

https://childmaltreatmentresearchblog.wordpress.com/about/

Emergent Cognitive Architecture

- Two different views of development
  - Individual
  - Social **(i.e., social factors in shaping cognitive development)**
- Two complementary theories of cognitive development

Jean Piaget
1896–1980

Lev Vygotsky
1896–1934

- **Stages of development**: **Piaget's theory** proposes that cognitive development occurs in four distinct stages. **Vygotsky's theory** emphasizes the continuous development of cognitive processes through social interaction.

- **Focus on individual vs. social factors**: **Piaget's theory** emphasizes the role of individual factors, such as biological maturation and experience, in shaping cognitive development. In contrast, **Vygotsky's theory** emphasizes the role of social and cultural factors, such as social interaction and language, in shaping cognitive development.

Emergent Cognitive Architecture

The cognitive architecture is the system's phylogenetic configuration

- The basis for ontogenesis: growth and development
  - Innate skills
  - Core knowledge (cf. E. Spelke)

- A structure in which to embed mechanisms for
  - Perception
  - Action
  - Adaptation
  - Anticipation
  - Motivation
  - ... Development of all these

Emergent Cognitive Architecture

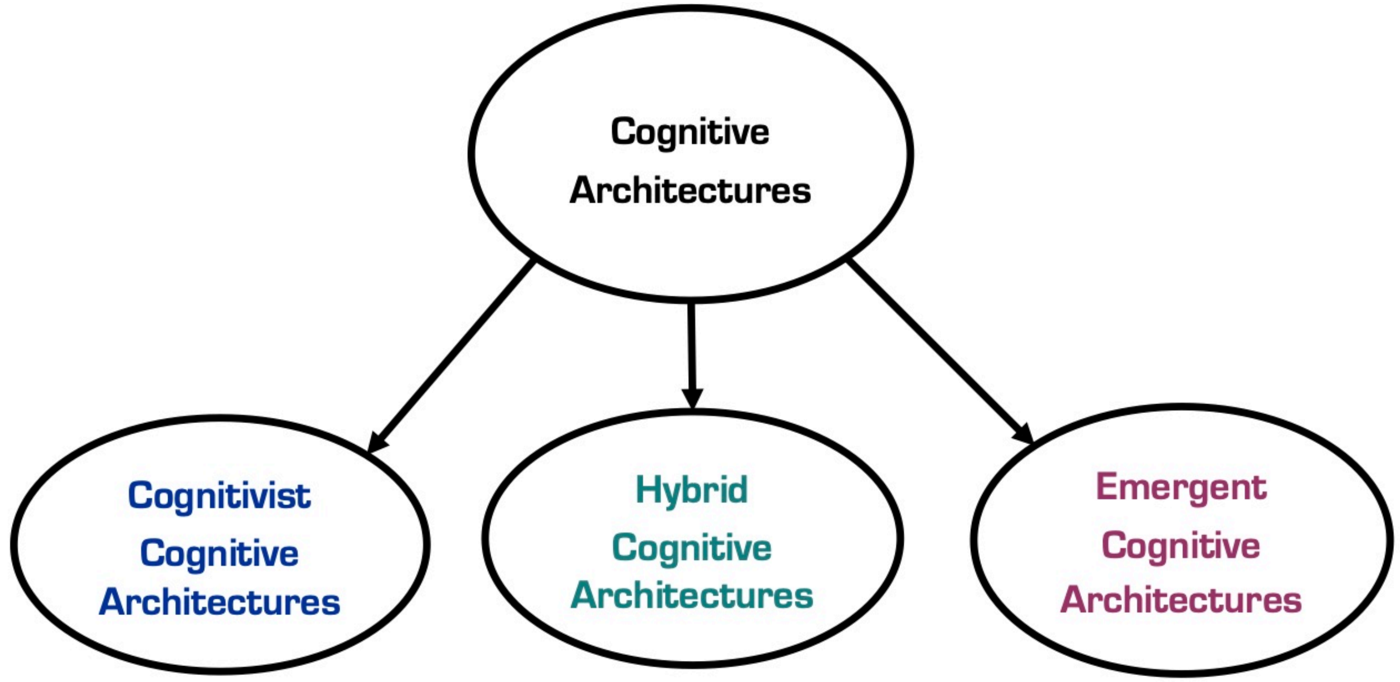
Strong focus on

- Autonomy-preserving, anticipatory, adaptive skill construction
- The morphology of the physical body in which the architecture is embedded

The emergent approach rejects:

- Dualism between mind and body
- Functionalism that treats cognitive mechanisms independently of the physical platform

Cognitive Architectures



Framework in which to embed knowledge

- Memories
- Formalisms for learning
- Programming mechanism

Phylogeny - basis for development

- Innate skills & core knowledge
- Memories
- Formalism for autonomy
- Formalism for development

- **Hybrid cognitive architectures** aim to combine the symbolic, rule-based processing of **cognitivist architectures** with the parallel, distributed processing of emergent architectures (i.e., *across many interconnected processing units, such as neurons or artificial nodes in a neural network*).
- For example, a **hybrid cognitive architecture** might use symbolic representations to encode abstract concepts and rules but also incorporate **neural networks** or other parallel processing techniques to support more flexible and adaptive behavior. The architecture might also include mechanisms for learning and adaptation, such as reinforcement learning or evolutionary algorithms.

Core Cognitive Abilities

1. Perception
2. Attention
3. Action selection
4. Memory
5. Learning
6. Reasoning
7. Metacognition
8. Prospection

I. Kotseruba and J. Tsotsos. 40 years of cognitive architectures: core cognitive abilities and practical applications. Artificial Intelligence Review, Vol. 53, No. 1, pp. 17-94, 2020.

Not included in [Kotseruba and Tsotsos 2020]

Core Cognitive Abilities

1. Perception
2. Attention
3. Action selection
4. Memory
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8. Prospection

Makes use of many sensory modalities, e.g. vision, audition, and haptic (tactile and kinesthetic)

Perception transforms raw input into the system's internal representation

Core Cognitive Abilities

1. Perception
2. Attention
3. Action selection
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8. Prospection

Reduces the information a cognitive system has to process by selecting relevant information & filtering out irrelevant information

Selective mechanisms

Choose one entity from many, e.g. gaze & viewpoint

Restrictive mechanisms

Choose some entities from many:

Priming what to look for or where to look for it

Suppressive mechanisms

Suppress some entities from many

i.e. features, objects, or locations that are not relevant

[Kotseruba and Tsotsos 2020]

Core Cognitive Abilities

1. Perception
2. Attention
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Determines what the agent should do next

Planning
Determines a sequence of steps to read a certain goal prior to execution of the plan

Dynamic action selection
Selection of one action based on knowledge at the time, typically using winner-take-all, probabilistic, or pre-defined order selection mechanisms

Core Cognitive Abilities

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Kotseruba and Tsotsos identify six types of memory

Short-term sensory memory	recent percepts
Short-term working memory	information relevant to current task
Long-term episodic memory	key to anticipation; autobiographical
Long-term semantic memory	general knowledge about the world
Long-term procedural memory	motor skills
Long-term global memory	for architectures that don't draw a type-duration distinction:

Core Cognitive Abilities

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The ability of a system to improve performance over time through the acquisition of knowledge or skill

Declarative learning …	explicit knowledge acquisition
Non-declarative learning …	perceptual, procedural, associative, non-associative learning

Supervised learning
Unsupervised learning
Reinforcement learning

- Examples of **declarative learning** include learning that "*if the goal is to travel, and you're in a car, then press the gas pedal*", and learning a math fact like "*7 × 8 = 56*".
- Examples of **non-declarative learning** include learning to ride a bike, developing motor skills for a sport, or learning to associate a certain smell with a particular emotion.

Core Cognitive Abilities

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- **Abduction**: Abductive reasoning is a type of reasoning in which the best possible explanation is sought for a given observation or set of observations. For example, if we observe that there is water on the floor, we might use abductive reasoning to conclude that the water was spilled, even though we did not observe someone spilling it.

The ability to logically and systematically process knowledge, typically to infer conclusions.

Three classical forms of logical inference: deduction, induction, abduction.

Reasoning focusses on the practical objective of finding the next (best) action to perform

- **Deduction**: Deductive reasoning is a type of logical reasoning in which a conclusion is drawn from one or more premises. For example, if we know that all cats are mammals and that Fluffy is a cat, then we can logically conclude that Fluffy is a mammal.

- **Induction**: Inductive reasoning is a type of reasoning in which generalizations are made based on observations or specific instances. For example, if we observe that every cat we have ever seen has been black and white, we might generalize that all cats are black and white, even though this may not be strictly true.

Core Cognitive Abilities

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Metacognition refers to the ability of a cognitive system has to monitor its internal cognitive processes, reason about them, and adapt them

Metacognition is needed for social cognition if the agent is to form a theory of mind, also known as perspective-taking, i.e. the ability to infer the cognitive states of other agents with which it is interacting

Core Cognitive Abilities

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Prospection – the capacity to anticipate the future – is one of the hallmark attributes of cognition

It lies at the heart of the other core characteristics of a cognitive agent: autonomy, perception, action, learning, and adaptation

[Vernon 2014]

It is central to action since actions are goal-directed and guided by prospective information

[von Hofsten 2009]

Internal simulation plays a key role in prospection

Summary

- **Today we discussed**

  1- Cognitive architectures

- **Next week**

  1- We will continue discussing the cognitive architectures